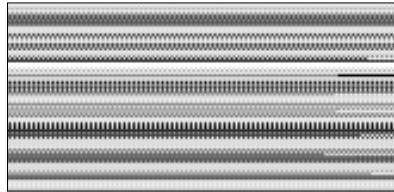


the last few chapters) but when it comes to organic, hand-crafted entities, terrains and shapes with more random patterns such as swimming pools or faces, it's Sandbox to the rescue.

5.1 Sandbox Tools

Sandbox tools are provided with SketchUp to enable the creation and modification of complicated and irregular surfaces.

It's typical of 3D-modelling software packages, including CAD, to facilitate terrain designing by dividing the surface you create into a mesh of triangles—small and big—which is, in terrain modelling lingo, called a TIN (Triangulated Irregular Network). The TIN is infinitely convenient as it helps you work on minute aspects of the terrain you're creating, by further sub-dividing the triangles into smaller triangles. These triangulated faces, however, when smoothed out, appear like a single smooth surface.



This is a TIN

If you find the triangles a bit too square, you could import or create a polygon mesh using other SketchUp tools and then proceed to work on it as you would with a TIN, although, manually creating them would be a bit tedious. However, SketchUp only creates triangular meshes with Sandbox.

5.1.1. Angling Triangles

When you take a look at the hidden geometry of a terrain you've created, you'll find that the triangles are oriented in different directions—sometimes horizontal and sometimes vertical. This orientation is called 'triangulation'. This triangulation can be modified. But before we enter that, let's figure out how a TIN can be created.

5.1.2. Hammering out TINs

You could create a triangulated surface by any of the following ways:

1. Using the "Sandbox From Contours Tool"